

Thesis Proposal Change Report

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Research Title: Lightweight Deep Learning IDS for Resource-Constrained IoT Devices

No.	Section	Change Made	Reason
1	Introduction	Revised discussion on IoT security challenges, deep learning-based IDS, and deployment limitations.	Improve clarity and strengthen research context.
2	Research Objectives	Refined objectives to align with lightweight IDS development, optimisation, and deployment on resource-constrained IoT devices.	Improve alignment between objectives and methodology.
3	Problem Statement	Rewritten to better define limitations of existing IDS solutions and highlight deployment challenges in IoT environments.	Address supervisor and panel feedback.
4	Literature Review (IDS Fundamentals)	Improved explanation of signature-based and anomaly-based intrusion detection approaches.	Enhance technical background and readability.
5	Literature Review (Deep Learning IDS)	Expanded discussion of CNN, LSTM, and deep learning-based intrusion detection methods.	Strengthen technical foundation of the study.
6	Literature Review (Lightweight Models)	Added TinyML, TinyIDS, MobileNet, and lightweight architectures.	Incorporate recent developments and current research trends.
7	Literature Review	Added recent references from 2024–2025.	Improve recency and relevance of literature.
8	Research Gap	Expanded deployment-focused evaluation and Raspberry Pi implementation discussion.	Clearly define research contribution.
9	Methodology	Revised methodology framework and workflow.	Improve methodological structure and clarity.
10	Data Collection and Preprocessing	Added dataset justification and preprocessing workflow.	Address dataset-related feedback.
11	Model Development	Added CNN justification and comparison with LSTM/CNN-LSTM.	Improve model selection rationale.
12	Lightweight	Added pruning, quantisation,	Strengthen

	Optimisation	TensorFlow Lite conversion and simplification.	deployment methodology.
13	Model Training and Evaluation	Expanded evaluation metrics and deployment measures.	Support practical deployment assessment.
14	Tools and Environment	Added TensorFlow Lite, Raspberry Pi deployment environment and libraries.	Improve implementation feasibility.
15	Proposed System Architecture	Updated architecture diagram and workflow.	Improve visual representation and technical clarity.
16	Expected Outcomes	Revised expected outcomes to focus on feasibility and efficiency.	Improve academic rigor.
17	Timeline	Updated project timeline (May–November 2026).	Align with revised plan.
18	References	Updated citations and added recent literature.	Improve academic quality.